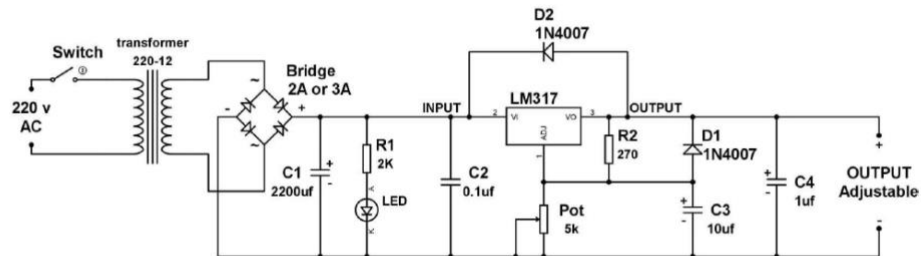
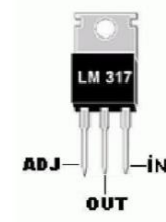


1- Adjustable Power Supply



Description:

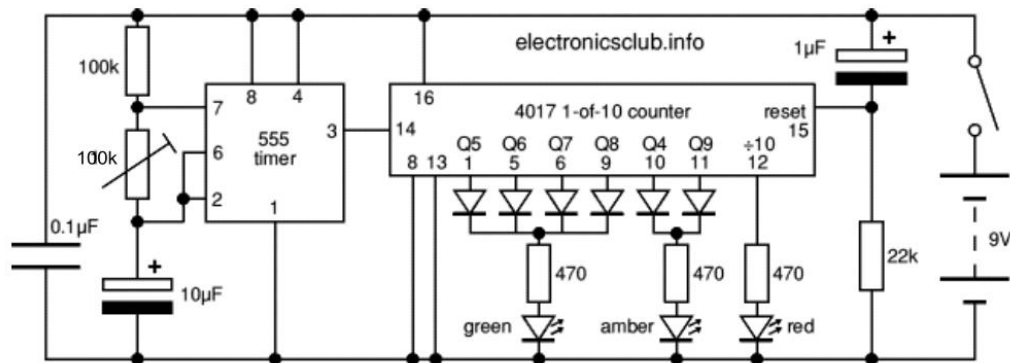
The LM317T is a fully adjustable 3-terminal positive voltage regulator capable of supplying 1.5 amps, with an output voltage ranging from around 1.25 volts to just over 30 volts. By using the ratio of two resistances, one of a fixed value and the other variable.



Components:

- LM 317
- 220/12 transformer
- Switch on/off
- Diode bridge 2A or 3A
- LED
- (2) Diode 1n 4007
- Cap 0.1 µF, 1 µF, 10 µF, 2200 µF
- Res. 270 Ω, 2 kΩ
- Potentiometer 5 kΩ
- PCB (1 layer)

2- Traffic light Circuit



Description:

This circuit uses IC 555 as an astable multi-vibrator. In astable mode the 555 timer puts out a continuous stream of clock pulses for the IC 4017. IC 4017 is a counter circuit, one output will be on at every clock pulse.

Components:

- IC 555
- IC 4017
- Switch on/off
- Battery 9 V
- LED (Green, Yellow, Red)
- Cap. 0.1 μ F, 1 μ F, 10 μ F
- Res. (3) 470 Ω , 22 k Ω , 100 k Ω
- Potentiometer 100 k Ω
- PCB (1 layer)

3- Ring Counter

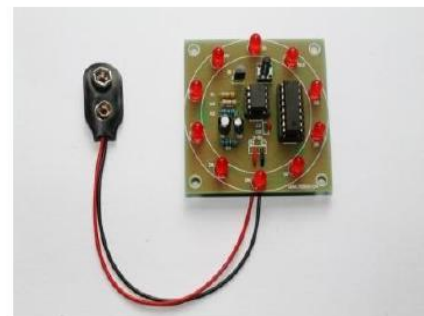
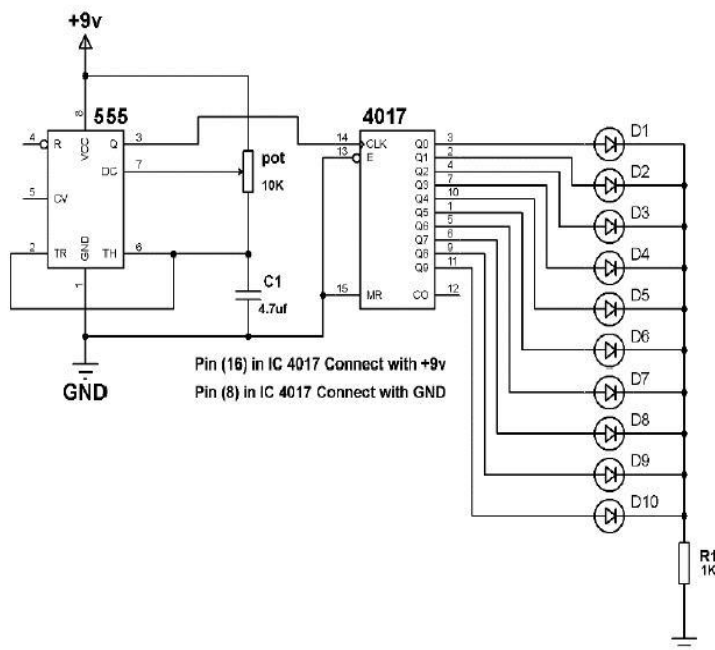
Description

This circuit uses NE 555 as an astable multivibrator. In a stable mode, the 555 timer puts out a continuous stream of clock pulses for the IC 2 CD 4017. In the CD 4017(IC 2) takes the clock pulses from the multivibrator as input and it provides the counters (decoded outputs). These counters are advanced by one during every positive cycle of the clock pulse if the Clock Inhibit signal is kept low.

Components:

- (10) led
- Res: 1k ohm
- IC 555
- IC 4017
- Cap: 4.7uf – two 0.1uf
- Potentiometer 10 k ohm

Schematic diagram:



4- BCD Adder

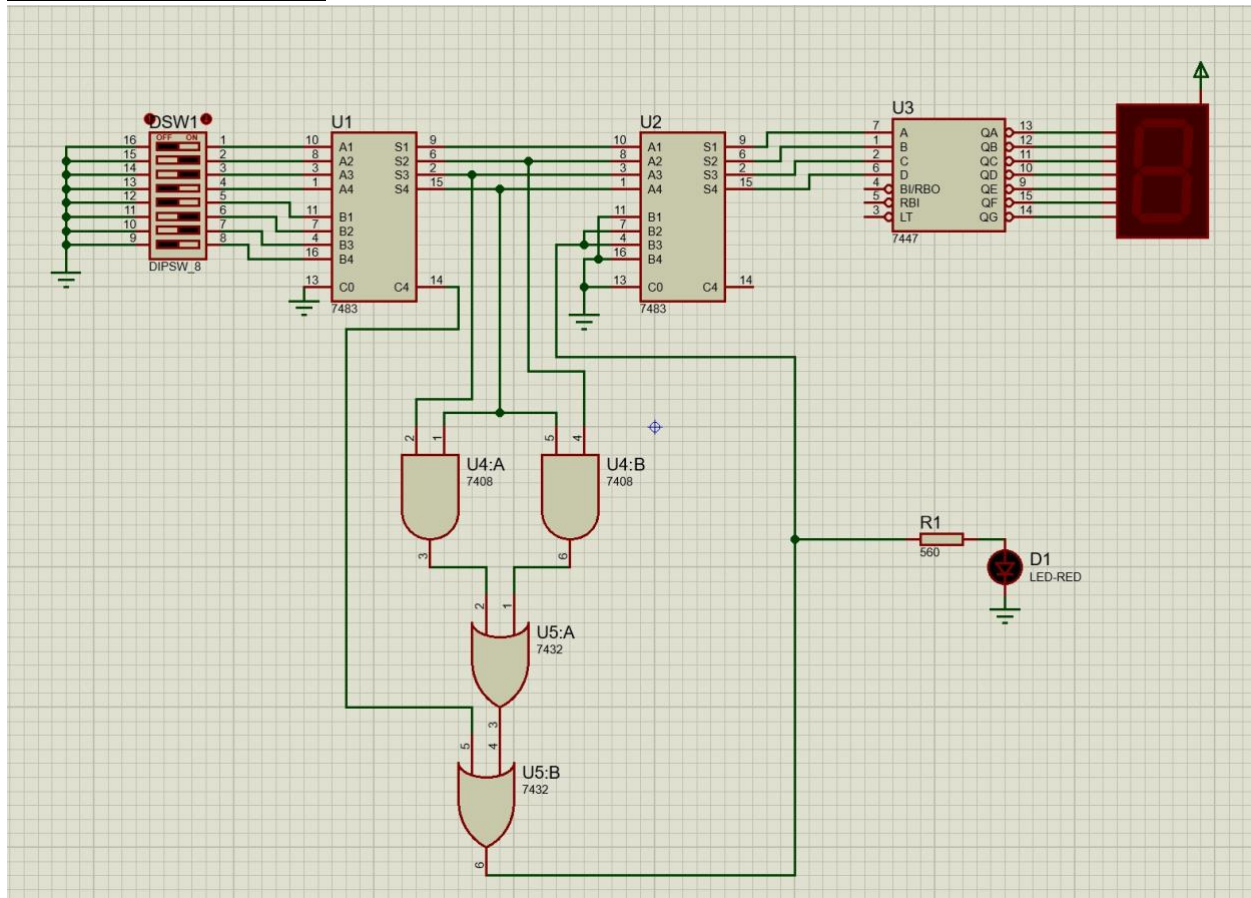
Description:

The Circuit adds two 4-bit numbers each number is binary represented to a decimal number from 0 to 9, and display the output on the decimal form on the 7-segment and a led which turns on only if the value of tens of the output equals to one.

Required Components:

- 1 - an IC of 7408 (4-AND Gates).
- 2- an IC of 7432 (4-OR Gates).
- 3 - 2 ICs of 7483 (4-bit Binary Adder).
- 4 - an IC of 7447 (7- Segment Common Anode Decoder).
- 5 - 7 segment Common Anode.
- 6 – LED.
- 7- Battery.
- 8 - DIP Switch - 8 Buttons.

Circuit Simulation:



5- Lighting Lamp Using Timer

Description:

In this circuit, we use 2n2222 as a switch to open a relay, which will control the AC current through the lamp. By using IC 555 as monostable multivibrator, the potentiometer value we will control the opening time of the relay as following:

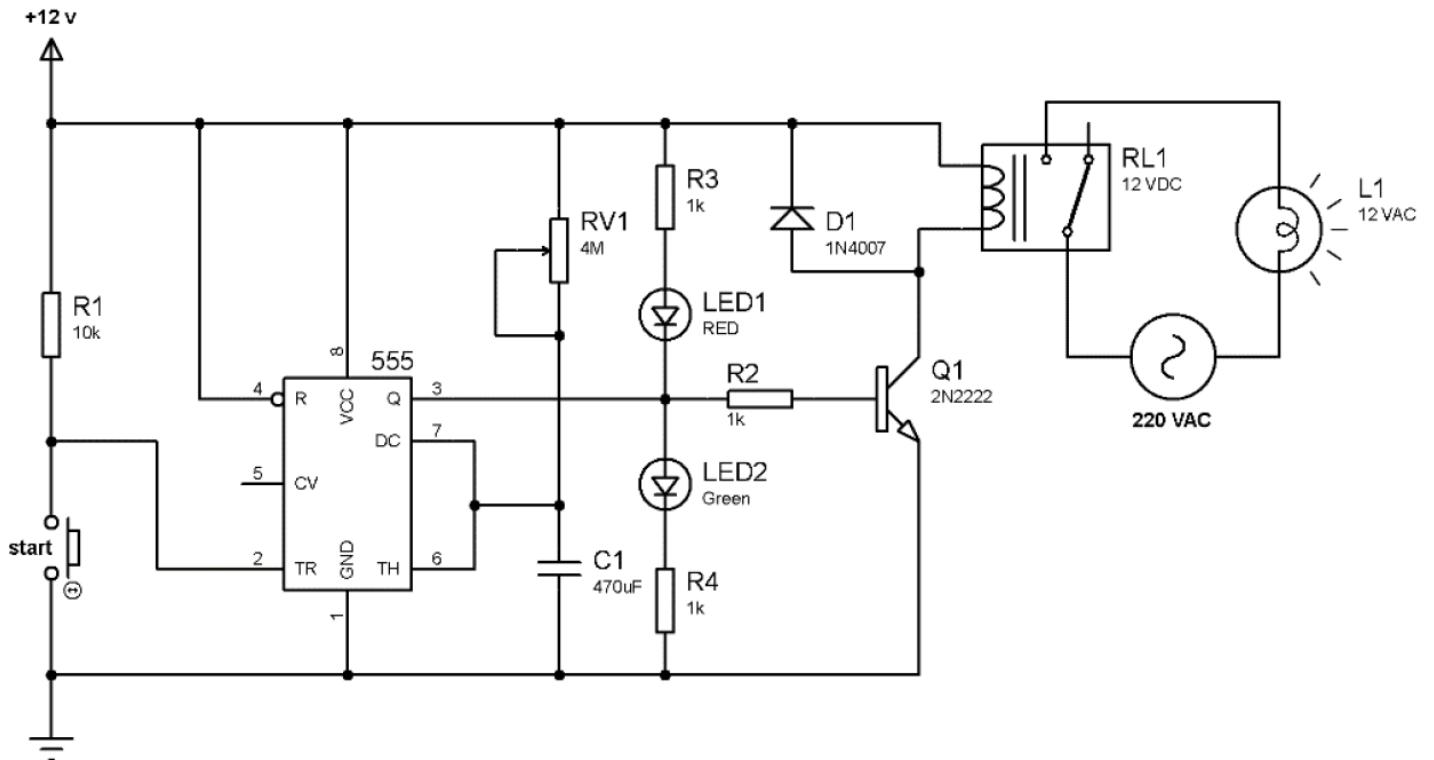
- If pot. value is 100K the lamb will be on for 1 min.
- If pot. value is 500K the lamb will be on for 5 min.
- If pot. value is 1.5M the lamb will be on for 15 min.
- If pot. value is 2M the lamb will be on for 30 min.
- If pot. value is 4M the lamb will be on for 60 min.

If you replace the Pot by fixed resistor, you will get the lamp on for a fixed time too.

Components:

- 12volt Battery
- 555 timer IC.
- Transistor 2n2222.
- Diode 1n4007.
- Relay 12VDC.
- Lamp 220VAC/25w.
- Electrical male socket with wire
- (1) push ON, push OFF button.
- TOP potentiometer 4Mohm.
- (3)Resistors - 1k.
- (2)LED red and green.
- (1)Resistors - 10k.
- Capacitor 470uF.
- Printed Circuit Board (pcb 1 layer).
- (3)screw terminal for pcb (two input)

Circuit Diagram:



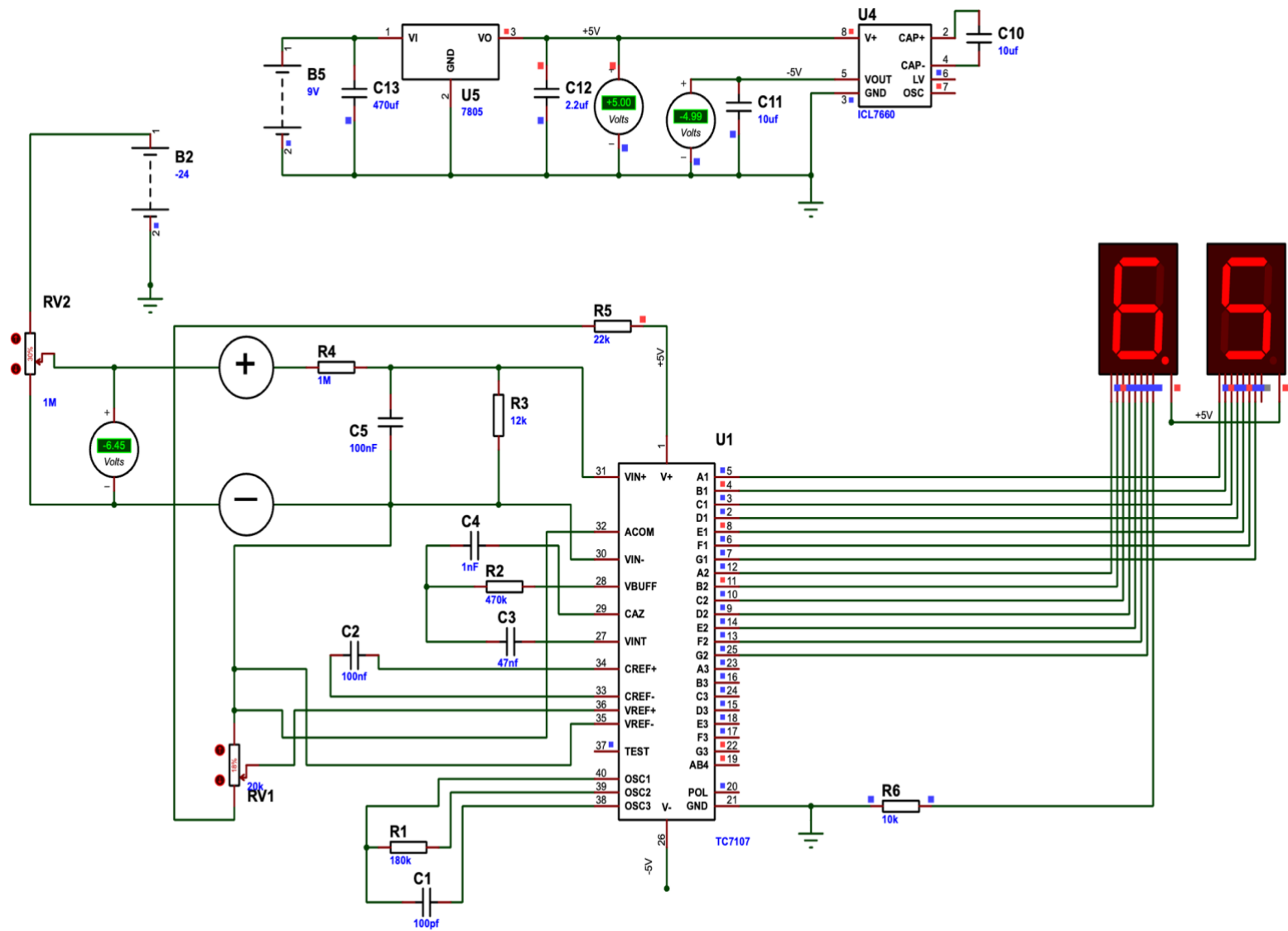
6- Digital Voltmeter

Description:

The following circuit uses ADC (Analog to Digital converter) and 7-Segment display to read and then display analog voltage with one floating point precision from 0.0 volt to 9.9 (if you used only two 7 Seg) or ± 99.9 Volt range (if you used four 7 Seg).

Components:

- 1- 9-volt Battery.
- 2- IC TC7107 or 7107 (ADC).
- 3- Two or four 7-Segment displays with Common anode.
- 4 - Pot: 20K
- 5 - Resistors:
180K – 470K – 12K – 1M – 22k – 10K
- 6 - Capacitors:
100pF – 100nF – 220nF – 47nF – two 10uF – 470uF – 2.2uF
- 7- Printed Circuit Board (PCB 1 layer).





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Full Circuit

